EIGEN · SUMMARY

Spectral properties at a glance

Nine operations, each linked to the section that derives it. The column worth watching is whether the eigenvectors survive — most operations move the eigenvalues while leaving the directions fixed, and the two exceptions are the informative ones.

09
OPERATIONS

UNCONDITIONAL 4

iv Powers A^k has eigenvalues λ_i^k — UNCONDITIONAL

REQUIRES k a positive integer

Eigenvectors unchanged. Apply A repeatedly to $\mathbf{v}$ and each pass multiplies by λ again. This is why $|\lambda| < 1$ means $A^k \to O$ and $|\lambda| > 1$ means it blows up — the whole of iterative stability sits in this line.

v Polynomials $q(A)$ has eigenvalues $q(\lambda_i)$ — UNCONDITIONAL

REQUIRES any polynomial q

Eigenvectors unchanged. Generalises the two entries above — powers and shifts are both special cases. It is also what makes the Cayley–Hamilton theorem believable: the characteristic polynomial sends every eigenvalue to zero.

vi Shift $A + cI$ has eigenvalues $\lambda_i + c$ — UNCONDITIONAL

REQUIRES any scalar c

Eigenvectors unchanged. Adding a multiple of the identity slides the whole spectrum along the number line, leaving the directions alone — which is what makes shifted power iteration and the shifted QR algorithm work.

vii Scalar multiple cA has eigenvalues $c\lambda_i$ — UNCONDITIONAL

REQUIRES any scalar c

Eigenvectors unchanged. Contrast the shift directly above: scaling stretches the spectrum about zero, shifting translates it. Neither disturbs the eigenvectors.

CONDITIONAL 5

i Trace $\operatorname{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$ — CONDITIONAL

HOLDS WHEN eigenvalues counted with algebraic multiplicity, over $\mathbb{C}$

Spectral information read off the diagonal without solving anything. Over $\mathbb{R}$ the sum can appear to fail because some eigenvalues live in $\mathbb{C}$ — a rotation has trace $2\cos\theta$ and no real eigenvalues at all. See trace properties.

ii Determinant $\det(A) = \lambda_1\lambda_2\cdots\lambda_n$ — CONDITIONAL

HOLDS WHEN with algebraic multiplicity, over $\mathbb{C}$

The product rather than the sum. It gives the invertibility criterion directly: A is invertible exactly when no eigenvalue is zero, since a single zero factor kills the whole product.

iii Inverse A^{-1} has eigenvalues $1/\lambda_i$ — CONDITIONAL

HOLDS WHEN A invertible — no $\lambda_i = 0$

Eigenvectors unchanged. From $A\mathbf{v} = \lambda\mathbf{v}$, multiply both sides by A^{-1}/λ and the same $\mathbf{v}$ comes back with the reciprocal. The condition is not a technicality: a zero eigenvalue is precisely what makes the inverse fail to exist.

viii Transpose A^{T} has the same eigenvalues as A — EIGENVECTORS DIFFER

HOLDS WHEN always for the eigenvalues
FAILS WHEN for the eigenvectors — generally different
COMMON ERROR assuming the eigenvectors carry over because the eigenvalues do

Same characteristic polynomial, since $\det(A^{\mathsf{T}} - \lambda I) = \det(A - \lambda I)$. But the eigenvectors of A^{T} are the left eigenvectors of A, a different set. One of only two entries here where the spectrum survives and the directions do not.

> **WITNESS**
> `A = [[1, 1], [0, 2]]` — eigenvalues 1, 2
> eigenvector for λ=1: A → (1, 0), Aᵀ → (1, -1)

ix Similarity $P^{-1}AP$ has the same eigenvalues, multiplicities included — EIGENVECTORS MAP

HOLDS WHEN P invertible

Both algebraic and geometric multiplicities are preserved, so the entire spectral picture is basis-independent. The eigenvectors are not lost but relabelled: $\mathbf{v}$ becomes $P^{-1}\mathbf{v}$. That is the whole content of similarity — the same transformation, described from a different basis.

The pattern: anything built from A by polynomials, inversion or shifting keeps the eigenvectors and transforms the eigenvalues by the same rule. Transposition and similarity are the exceptions — they preserve the spectrum but change or relabel the directions, which is exactly why diagonalization is a statement about a basis rather than about numbers.